Query Option/ Operator	Description	Official Sample
$select		https://services.odata.org/OData/OData.svc/Products?$select=Price,Name
		https://services.odata.org/OData/OData.svc/Categories?$select=Name,Products&$expand=Products
		https://services.odata.org/OData/OData.svc/Categories?$select=Name,Products&$expand=Products/Suppliers
$inlinecount		https://services.odata.org/OData/OData.svc/Products?$inlinecount=allpages
		https://services.odata.org/OData/OData.svc/Products?$inlinecount=allpages&$top=10&$filter=Price gt 200
$format		https://services.odata.org/OData/OData.svc/Products?$format=json

Instance-based Authorization

Two types of authorization are supported:

- Read: data contributors can read the tracked process fields that are granted Read authorization.
- Report: data contributors can report events that are granted Report authorization.

To enable the instance-based authorization, you need to:

- connect with your business partner (via the ***Invite Business Partners*** app or the ***Discover Business Partners*** app)
- for the tracked process user model field where you enter LBN ID, add **LBN#** as the prefix of its value to enable Read or Report authorization. For example, if the LBN ID is 10002424, you need to enter `LBN#10002424`.

Instance-based Authorization Check for Write Service

To check if a data contributor has Report authorization for an event, the GTT system will check if the event's `senderPartyId` is equal to the tracked process's `partyId` or to the LBN ID with the "LBN#" prefix. Not that events reported using **Provider API for Order Events (GTT)** are an exception, the GTT system won't do this check for these events.

For predefined event types, data contributors can only report GTTDelayedEvent and GTTOnTimeEvent.

Authorization for instances can be inherited and checked through correlation. For example, suppose tracked process A observes tracked process B and tracked process B observes tracked process C.



- If you have authorization for tracked process C, you will also have authorization for B and A. Then events reported to C are also correlated to B and A.

- If you don't have authorization for tracked process C, the system will check if you have authorization for B. If so, events will be reported to B and correlated to A. Otherwise, the system will check if you have authorization for A.
- If tracked process C doesn't exist, the system will check if you have authorization for B. If so, events will be reported to B and correlated to A. Otherwise, the system will check if you have authorization for A.

Instance-based Authorization Check for OData Read Service

1. Solution owners have Read authorization for their instances.
2. Data contributors can only query user-defined process entities, user-defined event entities, GTTDelayedEvent and GTTOnTimeEvent.
    - When querying process entities, you cannot filter/expand "QualifiedTrackingId" and user defined process entities.
    - When querying event entities, you cannot filter/expand core entities and user-defined process entities.
    - When querying any process entities, data will be filtered by process "partyId" and READ annotation specified LBN IDs. Note that at least one of process "partyId" and READ annotation specified LBN IDs should be equal to the user's LBN ID.
    - When querying any event entities, data will be filtered by "senderPartyId", "partyId" and READ annotation specified LBN IDs. Note that at least one of "senderPartyId", "partyId" and READ annotation specified LBN IDs should be equal to the user's LBN ID.

Notes

There are several points to note:

1. It's not supported that "/","//" ";", "%", "\" appear in the key of the query entity.

    Example

    Unsupported example:

    ```
    https://< Server Address
            >/api/outbound/odata/v1/com.gttsampleso01.gtt.app.sof.sofService/
            SalesOrderItem(id=guid'6d49b8f3-5599-5cbf-b9d0-b06bfb4e24a9', altkey='/;\%-1234567
    ```

2. It's recommended to keep the layer of the navigation or expand no more than 3 layers.

    Example

    Not suggested example:

    ```
    https://< Server Address
          >/api/outbound/odata/v1/com.gttsampleso01.gtt.app.sof.sofService/
          TrackedProcess(guid'6d49b8f3-5599-5cbf-b9d0-b06bfb4e24a9')?
          $expand=plannedEvents/lastProcessEventDirectory/event/references
    ```

3. The max limit of number of records in the response is 256.
4. For $expand, the use of a same entity name is not supported.

Example

Unsupported example:

```
https://< Server Address
        >/api/outbound/odata/v1/com.gttsampleso01.gtt.app.sof.sofService
        /InboundDeliveryItem(guid'323f6e3f-b3db-5969-b32f-3364ce2db76d')?$expand=freightUni
```

5. If the query result of '$expand' expression is greater than the allowed capacity limit of 200,000 records, a bad request response will be thrown. Your request will be rejected to avoid the system crash.

Location Service

Description

- technical name: GTT_V2_OData_Location
- purpose: to create or read locations
- protocol: OData
- users: technical. For more, see the "Create a Technical User and Assign Role Collection" section in ***Administration Guide for Solution Owners.***
- used to:
    - Create locations
    - Read locations

Create Locations

Request

Request URI

https://<server address>/api/location/v1/Location

HTTP Method

Post

Request Header

Pass additional information with the request.

Request Header Properties

Property	Type	Mandatory	Description
isCorrectGeocode	String	No	Value can be true or false. GeoCoordinate will be generated by location service based on the address data in request body if isCorrectGeocode is set to true.

Request Body

The location information.

Sample Code

```
{
    "ObjectTypeCode": "LogisticLocation",
    "SourceSystem": "ECC_001",
    "LogicalSystem": "ECCCLNT001",
    "ExternalId": "011",
    "Longitude": "12.12345000",
    "Latitude": "21.12345000",
    "LocationDescription": "Created by CoreEngine: 011",
    "AddressTimeZone": "AST",
    "CountryCode": "US",
    "RegionCode": "IA",
    "CityName": "Des Moines",
    "StreetName": "TIanze road",
    "HouseNumber": "No.10",
    "HouseNumberSupplementText": "1111",
    "PostalCode": "100000",
    "Number": "80016666",
    "NumberExtension": "123",
    "EmailAddress": "CORE_ENGINE_011@example.com",
    "Gln": "0234567890123",
    "Duns": "0234567890123",
    "Unlocode": "ggabc",
    "IataAirportCode": "abc"
}
```

Note

The value of the field “SourceSystem” should be the value that you configured in the Data Replication Framework in the SAP S/4HANA source system. For more, see the first procedure of the Managing Locations section in the guide Application Help.

The value of the field “LogicalSystem” should be the value that you configured in SAP ERP. For more, see the Define Logical Systems section in the guide How to Send Documents from SAP ERP to SAP Business Network Global Track and Trace.

Response

Response Body

The created location.

Read Locations

Request

Request URI

https://<server address>/api/location/v1/Location

Example

https://*<server address>*/api/location/v1/Location?$top=2&$filter=externalId eq '011'

HTTP Method

Get

Request Body

The request body is empty.

Response

Response Body

The locations satisfying the conditions in the query parameter.

Data Export

Description

- technical name: GTT_V2_HTTP_DATA_EXPORT
- purpose: to export the business data in the cloud environment
- protocol: HTTP
- users: technical. For more, see the "Create a Technical User and Assign Role Collection" section in ***Administration Guide for Solution Owners.***
- used to:
    - Create a data export request
    - Retrieve all data export requests
    - Retrieve one data export request by ID
    - Retrieve files that contain the exported data

Note: To use this interface, you must have the `MetadataServiceIntegrationTemplate` role in your role collection. For more, see the "Available Roles for Solution Owners" section in the ***Administration Guide for Solution Owners***.

Create Data Export Request

Request

Request URI

https://server address/api/data-load/v1/{model namespace}/exportBusinessData

Example

https:// *<server address>*/api/data-load/v1/com.lbngttapps.gtt.app.gttft1/exportBusinessData

HTTP Method

POST

Request Body

Sample Code

```
{
    "startTime": "2023-11-15T08:30:16.123Z",
    "endTime": "2023-12-06T08:30:21.456Z",
    "queryByField": "LAST_CHANGE_DATETIME",
    "processTypes": [
        "com.lbngttapps.gtt.app.gttft1.tp1.tp1",
        "com.lbngttapps.gtt.app.gttft1.tp2.tp2"
    ]
}
```

Response

Sample Code

```
{
    "id": "3854cf7f-88ec-47fc-8329-932f9811bada",
    "startTime": "2023-11-15T08:30:16.123Z",
    "endTime": "2023-12-06T08:30:21.456Z",
    "status": "CREATED",
    "receivedAt": "2023-12-07T08:30:21.145Z",
    "requestedBy": "user@sap.com",
    "modelNamespace": "com.lbngttapps.gtt.app.gttft1",
    "queryByField": "LAST_CHANGE_DATETIME",
    "processTypes":[
        "com.gttsampleso01.gtt.app.gttft1.tp1.tp1",
        "com.gttsampleso01.gtt.app.gttft1.tp2.tp2"
    ]
}
```

Retrieve All Data Export Requests

Request

Request URI

https://server address/api/data-load/v1/{model namespace}/data-export

Example

https://*<server address>*/api/data-load/v1/com.lbngttapps.gtt.app.gttft1/data-export

HTTP Method

GET

Request Body

The request body is empty.

Response

Sample Code


```
[
    {
        "id": "3854cf7f-88ec-47fc-8329-932f9811bada",
        "startTime": "2023-12-01T07:08:06.123Z",
        "endTime": "2023-12-17T07:08:11.234Z",
        "status": "COMPLETED",
        "receivedAt": "2023-12-18T08:30:21.145Z",
        "requestedBy": "user@sap.com",
        "modelNamespace": "com.lbngttapps.gtt.app.gttft1",
        "queryByField": "LAST_CHANGE_DATETIME",
        "fileInfoList": [
            {
                "fileID": "N01aqwJizZye4X7HeHS01TCb7CRdX8U6xUffMjN-Fls",
                "fileName": "tp2_20231218083335.zip",
                "creationDateTime": "2023-12-18T08:33:35.145Z"
            },
            {
                "fileID": "LswV5M5-ngwYO6_n0nVIV8zyIVq7pPB-gFgE95xu93U",
                "fileName": "tp1_20231218083335.zip",
                "creationDateTime": "2023-12-18T08:33:35.145Z"
            }
        ]
    }
]
```


Retrieve One Data Export Request by ID

Request

Request URI

https://server address/api/data-load/v1/{model namespace}/{data export ID}

Example

https://*<server address>*/api/data-load/v1/com.lbngttapps.gtt.app.gttft1/data-export/3854cf7f-88ec-47fc-8329-932f9811bada

HTTP Method

GET

Request Body

The request body is empty.

Response

The following JSON response structure is returned:

Sample Code

```
{
    "id": "3854cf7f-88ec-47fc-8329-932f9811bada",
    "startTime": "2023-12-01T07:08:06.123Z",
    "endTime": "2023-12-17T07:08:11.234Z",
    "status": "COMPLETED",
    "receivedAt": "2023-12-18T08:30:21.145Z",
    "requestedBy": "user@sap.com",
    "modelNamespace": "com.lbngttapps.gtt.app.gttft1",
    "queryByField": "LAST_CHANGE_DATETIME",
    "fileInfoList": [
        {
            "fileID": "N01aqwJizZye4X7HeHS01TCb7CRdX8U6xUffMjN-Fls",
            "fileName": "tp2_20231218083335.zip",
            "creationDateTime": "2023-12-18T08:33:35.145Z"
        },
        {
            "fileID": "LswV5M5-ngwYO6_n0nVIV8zyIVq7pPB-gFgE95xu93U",
            "fileName": "tp1_20231218083335.zip",
            "creationDateTime": "2023-12-18T08:33:35.145Z"
        }
    ]
}
```

Retrieve Files That Contain the Exported Data

Request

Request URI

https://<server address>/api/outbound/v1/document/{file ID}

Example

https://*<server address>*/api/outbound/v1/document/N01aqwJizZye4X7HeHS01TCb7CRdX8U6xUffMjN-Fls

HTTP Method

Get

Response

File **tp2_20231218083335.zip**

SOAP Message Interfaces

This section introduces SOAP message interfaces used by SAP Business Network Global Track and Trace.

TransportationOrderGenericTrackedProcessRequest_Out

As a solution owner, you can create a freight order in TM embedded in SAP S/4HANA and send it for tracking in SAP Business Network for Logistics.

Technical Name

`TransportationOrderGenericTrackedProcessRequest_Out`

Description

You can use this inbound service provided by SAP Business Network for Logistics to create a freight order for tracking from your external system, such as TM embedded in SAP S/4HANA. This will enable you to track a freight order. For detailed configurations, see Using SOAP Messages.

Frequently Asked Questions

This section answers the questions that are frequently asked when you are using the tracking APIs.

As a solution owner, can I request a new event type in the Provider API for Order Events (GTT)?

Yes. You can report an incident on SAP ONE Support Launchpad categorized by component SBN-LBN-GTT to create a new event type. But before reporting the incident, please take a moment to familiarize yourself with the information below:

1. An event type means an action, such as Arrival or Departure. So it should not include any location-related information.
2. Location information could be maintained based on the stop ID (which is provided via the **Consumer API for Order Tracking (GTT)**), UNLOCODE, or IATACode.

If you want to add an Arrival or Destination event, you can use the `ARRIV_DEST` event for both of these events. The exact location should then be maintained in the location ID.

To foster standardization, please map your events to one of the existing event types. For a complete list of existing event types, see Event Code Descriptions. If you are unsure about which event type you should map with your event, you may also consider mapping it to the `OTHER` event type and adding a specific description for it.

Does SAP Business Network Global Track and Trace support geo location tracking for direct integration with carriers (no visibility provider in between)?

Yes. Carriers can also provide geo location updates via the **Provider API for Order Events (GTT)**.

Note that any event type can contain geo coordinates, such as Arrival, Departure, On Time, Delay, and so on.

As a solution owner, can I receive attachments from the data contributor as part of the Provider API for Order Events (GTT)?

Yes. You have two options to achieve that.

Option 1: The data contributor needs to maintain the `fileName`, `mimeType`, and `fileURL` that is publicly accessible. The GTT system will save the content from the URL provided as an attachment with the specified event.

Option 2: The data contributor needs to maintain the `fileName` and `fileContentBinaryObject` to upload a Base64-encoded attachment.

What is the size limit for attachments such as the proof of delivery that are sent by the data contributor?

The total size of the attachment(s) in an event may not exceed 10 MB.

As a solution owner, can I send a tracking request to multiple business partners regarding one specific document?

Yes. You can achieve this in the Manage Models app by using the event-to-action function `sendTrackingRequest`. For more, see the corresponding part in the section Event to Action External Functions of *Guide for Model Administrators*.